Naïve Bayes Classification



Anubha Gupta, PhD.
Professor
SBILab, Dept. of ECE,
IIIT-Delhi, India
Contact: anubha@iiitd.ac.in; Lab: http://sbilab.iiitd.edu.in

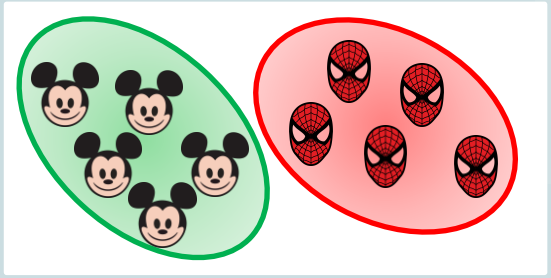
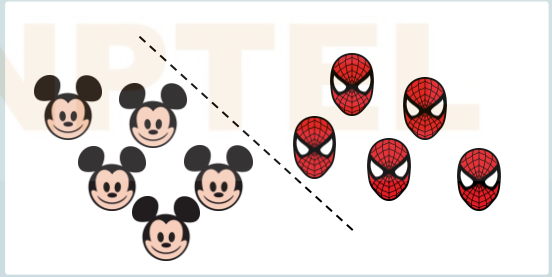


INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY **DELHI**

Naïve Bayes Classification

NPTEL

Motivation

Generative Classifier	Discriminative Classifier
Models the joint probability distribution of the input features given the class labels ($P(\mathbf{X}\|Y)$)	Models the conditional probability distribution of the class labels given the input features ($P(Y\|\mathbf{X})$)
Learns the characteristics of each class by estimating the probability distributions of the features within each class	Focuses on learning the decision boundary or the mapping function that directly separates the different classes.
	

Motivation

Generative Classifier	Discriminative Classifier
Can be used to generate synthetic samples that resemble the training data	It cannot be used to generate synthetic samples.
Can handle missing data	Do not inherently handle missing values
Examples: Naïve Bayes, Gaussian Mixture Models (GMM), and Hidden Markov Models (HMM).	Examples: Logistic Regression, Support Vector Machines (SVM), and Neural Networks.

We are going to study Naïve Bayes classifiers in this lecture.

NPTEL

Learning Objectives

In this module, we will learn about the Naïve Bayes classifier and study an example.

NPTEL

Naïve Bayes Classification

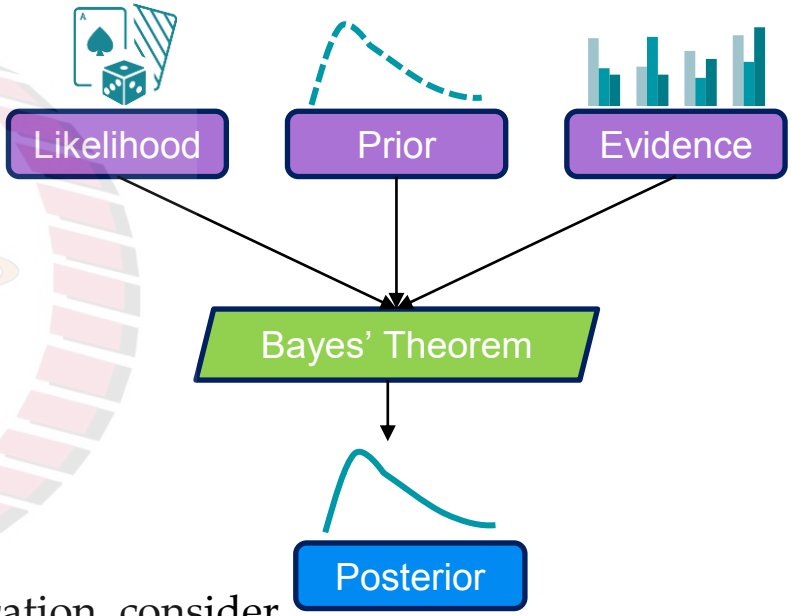
- A popular classification algorithm that is simple and easy to implement
- Based on Bayes' theorem
- **Assumption**: Features are conditionally independent given the class label, i.e., the presence or absence of one feature does not affect the presence or absence of another feature
- Features can be categorical or continuous.
- Considerably a fast algorithm



NPTEL

Bayes' Theorem-(Foundation of Naïve Bayes Classification)

"The probability of an event A given that event B has occurred is equal to the probability of event B given that event A has occurred, multiplied by the probability of event A occurring, and divided by the probability of event B occurring."



$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Likelihood Prior (a priori probability)

$$P(\theta|R) = \frac{P(R|\theta) \cdot P(\theta)}{P(R)}$$

Posterior

Evidence/ Data

In the context of classification, consider θ=y=class label, R=**x**= feature vector

$$P(y|\mathbf{x}) = \frac{P(\mathbf{x}|y) \cdot P(y)}{P(\mathbf{x})}$$

Assumptions

- Each feature is independent of all other features given the class. This is known as the "Naïve" assumption since it is often untrue in practice
- All features are equally important in predicting the class label
- Class distribution is known or can be estimated from the training data
- There are no missing values in the training data. If there are any missing values, they need to be imputed before training the classifier.
- Class labels are mutually exclusive. Each sample can only belong to one class label.

NPTEL

Types

Based on the type of data being used, Naïve Bayes classifiers are of three types.

1. **Gaussian Naïve Bayes**:
   - Assumes that the feature values are continuous & normally distributed
2. **Bernoulli Naïve Bayes**:
   - Assumes that the feature values are discrete with binary or Boolean values such as yes/no, true/false, etc.
   - **Application**: A particular word is present in a document or not.
3. **Multinomial Naïve Bayes**:
   - Assumes that the feature values are discrete or frequency-based
   - **Application**: Used for documentation classification tasks.

   Features are the frequency of words converted from a document.

NPTEL

Example of Naïve Bayes

$p(y|\mathbf{x}) = \frac{p(\mathbf{x}|y)P(y)}{p(\mathbf{x})}$

Consider the following vector of attributes: x_1 = likes Dosa; x_2 = likes Pizza; x_3 = drinks whiskey; x_4 = eats porridge; x_5 = likes to watch football

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

European

	S1	S2	S3	S4	S5	S6
x_1	0	1	1	1	0	0
x_2	0	0	1	1	1	0
x_3	1	1	0	0	0	0
x_4	1	1	0	0	0	1
x_5	1	0	1	0	1	0

Indian

	S1	S2	S3	S4	S5	S6	S7
x_1	1	1	1	1	1	1	1
x_2	0	1	1	1	1	0	0
x_3	0	0	1	0	0	1	1
x_4	1	0	1	1	1	1	0
x_5	1	1	0	0	1	0	0

$y = 0$ or 1, where 0 denotes a European and 1 denotes an Indian

$p(\mathbf{x}|y) = p(x_1|y)p(x_2|y)\ p(x_3|y)\ p(x_4|y)\ p(x_5|y)$

$p(x_1|y=\varepsilon)$

Independence assumptions of Naïve Bayes classifier

Example of Naïve Bayes

European- Class 0

	S1	S2	S3	S4	S5	S6
x_1	0	1	1	1	0	0
x_2	0	0	1	1	1	0
x_3	1	1	0	0	0	0
x_4	1	1	0	0	0	1
x_5	1	0	1	0	1	0

Indian- Class 1

	S1	S2	S3	S4	S5	S6	S7
x_1	1	1	1	1	1	1	1
x_2	0	1	1	1	1	0	0
x_3	0	0	1	0	0	1	1
x_4	1	0	1	1	1	1	0
x_5	1	1	0	0	1	0	0

$$p(y=0) = \frac{6}{13} \qquad p(y=1) = \frac{7}{13}$$

$$p(x_1 = 1 \mid y=0) = \frac{1}{2} \qquad p(x_2 = 1 \mid y=0) = \frac{1}{2} \qquad p(x_3 = 1 \mid y=0) = \frac{1}{3} \qquad p(x_4 = 1 \mid y=0) = \frac{1}{2}$$

$$p(x_5 = 1 \mid y=0) = \frac{1}{2} \qquad p(x_1 = 1 \mid y=1) = 1 \qquad p(x_2 = 1 \mid y=1) = \frac{4}{7}$$

$$p(x_3 = 1 \mid y=1) = \frac{3}{7} \qquad p(x_4 = 1 \mid y=1) = \frac{5}{7} \qquad p(x_5 = 1 \mid y=1) = \frac{3}{7}$$

Example of Naïve Bayes

Q: Given: $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, compute the country of the given person.

Solution: Let us compute the probability $p(y = 1|\mathbf{x})$.

Use Bayes Rules:

$$p(y|\mathbf{x}) = \frac{p(\mathbf{x}|y)p(y)}{p(\mathbf{x})}$$

Or

$$p(y = 1|\mathbf{x}) = \frac{p(\mathbf{x}|y = 1)p(y = 1)}{p(\mathbf{x}|y = 1)p(y = 1) + p(\mathbf{x}|y = 0)p(y = 0)}$$

Example of Naïve Bayes

$$p(y=1|\mathbf{x}) = \frac{p(\mathbf{x}|y=1)p(y=1)}{p(\mathbf{x}|y=1)p(y=1) + p(\mathbf{x}|y=0)p(y=0)} = \frac{a}{a+b}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$b = p(x_1 = 1|y=0)p(x_2 = 0|y=0)p(x_3 = 1|y=0)\, p(x_4 = 1|y=0)p(x_5 = 0|y=0)p(y=0)$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2} \times \frac{6}{13}$$

$$a = p(x_1 = 1|y=1)p(x_2 = 0|y=1)p(x_3 = 1|y=1)\, p(x_4 = 1|y=1)p(x_5 = 0|y=1)p(y=1)$$

$$= 1 \times \frac{3}{7} \times \frac{3}{7} \times \frac{5}{7} \times \frac{4}{7} \times \frac{7}{13}$$

$$p(y=1|\mathbf{x}) = \frac{a}{a+b} = 0.8076$$

$$p(y=0) = \frac{6}{13} \qquad p(y=1) = \frac{7}{13}$$

$p(x_1 = 1\|y=0) = \frac{1}{2}$	$p(x_1 = 1\|y=1) = 1$
$p(x_2 = 1\|y=0) = \frac{1}{2}$	$p(x_2 = 1\|y=1) = \frac{4}{7}$
$p(x_3 = 1\|y=0) = \frac{1}{3}$	$p(x_3 = 1\|y=1) = \frac{3}{7}$
$p(x_4 = 1\|y=0) = \frac{1}{2}$	$p(x_4 = 1\|y=1) = \frac{5}{7}$
$p(x_5 = 1\|y=0) = \frac{1}{2}$	$p(x_5 = 1\|y=1) = \frac{3}{7}$

NPTEL

Working

- **Process**:
  - Calculate prior probability of each class based on class frequencies in the training data
  - Estimate conditional probability of each feature given the class

- **Making Predictions**: For a new test sample,
  - Calculate the probability of each class given the input features using Bayes' theorem
  - Select the class with the highest probability as the predicted class

NPTEL

Advantages and Limitations

- **Advantages**:
  - **Simplicity**: Easy to implement and understand
  - **Efficiency**: Fast training and prediction times, suitable for real-time applications
  - Handles high-dimensional data well
  - Robust to irrelevant features

- **Limitations and Considerations**:
  - The independence assumption may not hold true in all datasets
  - Sensitivity to feature interactions
  - Requires a sufficient amount of training data to estimate reliable probabilities
  - Performance can be affected by imbalanced class distributions

NPTEL

Applications

- **Email Spam Filtering**: To classify emails as spam or not spam based on the content of the email
- **Sentiment Analysis**: To determine the sentiment of a piece of text (positive, negative, or neutral) based on the words used in the text
- **Text Classification**: To classify text into different categories based on the content of the text, such as news articles, product reviews, or social media posts
- **Recommendation Systems**: To recommend products or services to users based on their past behavior or preferences
- **Fraud Detection**: To detect fraudulent transactions or activities based on patterns in the data

In general, any classification task can be handled.

NPTEL

To Summarize

In this module, we studied Naïve Bayes Classifier.

NPTEL